भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 306: Machine Learning Lab

Evaluation Assignment 2

Instructions: This assignment is for evaluation and marks will be awarded. You need to complete execution by 12 PM. The saved results folder (consisting .csv files, plots, word file) has to be submitted through a Google form, which will be shared by Teaching Assistant.

1. Consider a scenario, where a financial institution uses the following features to predict whether a transaction is fraudulent:

   - Transaction amount
   - Account balance
   - Total Number of transactions
   - Transaction frequency (per month/ per day)
   - Time since last transaction

   Hint: The features have very different numerical scales. Perform the following tasks:

   (a) Generate a synthetic dataset to predict fraudulent/ non-fraudulent transactions by considering the above predictors/ features for 500 patterns.

   (b) Implement binary logistic regression model to design the hypothesis ($\hat{y}$) using Stochastic gradient Descent and 5-fold cross validation, without feature scaling. Consider the hyperparameters as below:
   - learning rate, α: $\{0.01\}$
   - ρ: $\{0.001\}$ where, ρ signifies absolute error-difference in two consecutive epochs. Consider ρ as the stopping criteria.

   (c) Re-implement 1(b) using Z-score normalization and compare the results. Report your observations.
   Hint: Ensure that the feature scaler is fitted only on the training set of each fold, and then use the fitted scaler to transform the test set.

   (d) Generate the confusion matrices for both the cases (1(b), 1(c)), and report the overall accuracy, class-wise accuracy, precision and recall, for all the 5-folds. Report the results separately for both the cases.

   (e) Build two logistic models with the following predictor settings:
   - Model A: Transaction amount & Transaction frequency (per month/ per day)
   - Model B: All the available predictors.

   Use the same hyper parameter settings as 1(b) and 5-fold cross validation.

   (f) Calculate the confusion matrix and mean-accuracy, mean-precision and mean-recall values for Model A and Model B, separately.

   (g) Report your observation on which model provides best predictive performance, and write down your result analysis in the word document from the results and plots/graphs.